

IEAP Series 02 – Data mining and statistical tests

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GitHub repository: <https://github.com/sdurand875-spec/IEAP-Series02-RStudio>

1 What is a statistical test, and when to use it?

1.1 What is a median? A mean?

The mean is the arithmetic average of a dataset. The median is the middle value when the data is sorted in order (Source: Google Gemini, see References)

1.2 What is a variance? A standard deviation?

Variance is the average squared deviation of each data point from the mean.

The standard deviation is the square root of the variance, which expresses the dispersion of the values. (Source: Jamet, Introduction aux statistiques, Licence STAPS Marseille)

1.3 What is a normal distribution?

A normal distribution is a continuous and symmetric. There is the same % in the 2 sides of the mean. And the mean = the median = mode. It is a bell-shaped probability distribution where data clusters around the mean, with 68% of values within one standard deviation and 95% within two. (Source: Jamet, Introduction aux statistiques, Licence STAPS Marseille)

1.4 What is a statistical test?

A statistical test uses sample data to test a hypothesis and draw conclusions about a larger population. (Source: Google Gemini, see References)

1.5 When to use a statistical test?

We use a statistical test to check if the results in your data are real, or if they just happened by chance. (Source: Google Gemini, see References)

1.6 What is a parametric test? A non-parametric test?

A parametric test assumes that your data follows a specific pattern (usually a normal distribution), while a non-parametric test doesn't make any assumptions about how the data is distributed. (Source: Jamet, Introduction aux statistiques, Licence STAPS Marseille)

1.7 What are the assumptions of parametric tests?

Common assumptions include normal distribution, equal variances between groups, independent observations, and data measured on an interval or ratio scale. (Source: Jamet, Introduction aux statistiques, Licence STAPS Marseille)

1.8 What are the assumptions of non-parametric tests?

Non-parametric tests assume that observations are independent, data is at least ordinal, and for tests like Mann-Whitney U, that both groups have a similar distribution shape. (Source: Google Gemini, see References)

1.9 What is a p-value?

A p-value is the probability of obtaining test results at least as extreme as the observed results, assuming the null hypothesis is true. (Source: Jamet, Introduction aux statistiques, Licence STAPS Marseille)

1.10 What is the risk of error when using a statistical test?

The primary risks are a Type I error alpha, which is rejecting a true null hypothesis (false positive), and a Type II error beta, which is failing to reject a false null hypothesis (false negative). (Source: Jamet, Introduction aux statistiques, Licence STAPS Marseille)

1.11 What is the difference between a paired and an unpaired test?

A paired test compares two set of measurements from the same group (like before and after a treatment), while an unpaired test compares two completely independent groups. (Source: Jamet, Introduction aux statistiques, Licence STAPS Marseille)

2 Effect of treatment over time

2.1 The data

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(here)
```

here() starts at /Users/sarahdurand/Desktop/M1 IEAP/Python-R-Git/Series02/Rstudio-Series02/IEAP-Series02-RStudio

```
prepost <- read.csv(here("data", "PrePost.csv"))
str(prepost)
```

```
'data.frame':  16 obs. of  2 variables:
 $ perf: int  125 130 132 135 136 138 140 145 127 132 ...
 $ time: chr  "Before" "Before" "Before" "Before" ...
```

```
summary(prepost)
```

	perf		time
Min.	:125.0	Length	:16
1st Qu.:	:132.0	N.unique	: 2
Median	:136.0	N.blank	: 0
Mean	:136.4	Min.nchar	: 5
3rd Qu.:	:140.2	Max.nchar	: 6
Max.	:148.0		

```
table(prepost$time)
```

After	Before
8	8

The file `PrePost.csv` contains 16 observations and 2 variables: a performance measure (`perf`) and the time of measurement (`time`), with 8 measurements before and 8 measurements after the rehabilitation program.

No variable identifies the participants, so the file itself does not tell us whether the same individuals were measured before and after.

Variable	Description	Unit	Possible values	Expected change
perf	Physical performance measured before and after the rehabilitation program	Not specified	Positive numerical values; observed range: 125–148	Expected to increase after rehabilitation
time	Time of measurement relative to the rehabilitation program	Not applicable	"Before", "After"	/

2.2 The analysis

2.2.1 Does the treatment have an effect?

The objective is to determine whether the rehabilitation program has an effect on physical performance. Because rehabilitation is expected to improve the patients' performance, we expect the performance values to be higher **after** the rehabilitation program than **before**. The comparison of interest is therefore: **Before vs After rehabilitation**.

```
# descriptive statistics for each time point
prepost %>%
  mutate(time = factor(time, levels = c("Before", "After"))) %>%
  group_by(time) %>%
  summarise(
    n      = n(),
    mean   = mean(perf),
    sd     = sd(perf),
    median = median(perf),
    IQR    = IQR(perf),
    min    = min(perf),
    max    = max(perf)
  )
```

```
# A tibble: 2 x 8
  time      n mean   sd median  IQR   min   max
<fct> <int> <dbl> <dbl> <dbl> <dbl> <int> <int>
1 Before     8 135.  6.20  136.    7   125   145
2 After      8 138.  7.01  138.   9.25  127   148
```

```
# difference between after and before
mean(prepost$perf[prepost$time == "After"]) -
  mean(prepost$perf[prepost$time == "Before"])
```

```
[1] 2.5
```

```
median(prepost$perf[prepost$time == "After"]) -
  median(prepost$perf[prepost$time == "Before"])
```

```
[1] 2
```

Descriptively, performance was slightly higher after rehabilitation than before, both in mean (137.6 ± 7.0 vs 135.1 ± 6.2 , i.e. +2.5 units) and in median (137.5 [IQR = 9.25] vs 135.5 [IQR = 7.00], i.e. +2 units). Mean $\pm$ SD is appropriate only if the data are normally distributed; otherwise, the median and IQR should be preferred. Normality is checked in question 3. In both cases, the difference is small compared with the variability between individuals, and the two distributions largely overlap (range 125–145 before vs 127–148 after). Whether this difference is statistically significant is tested in question 5.

2.2.2 What comparison are you going to make?

We compare physical performance **before** vs **after** the rehabilitation program (8 observations in each group).

Ideally, a before–after study uses a **paired comparison**, because the same patients are measured twice. The analysis is then based on the difference computed for each participant:

$$\text{Difference} = \text{After} - \text{Before}$$

However, the dataset does not contain any participant identifier, so it is impossible to know which “Before” value corresponds to which “After” value.

```
before <- prepost$perf[prepost$time == "Before"]
after  <- prepost$perf[prepost$time == "After"]

before
```

```
[1] 125 130 132 135 136 138 140 145
```

```
after
```

```
[1] 127 132 133 136 139 141 145 148
```

```
# are the values sorted in increasing order within each group?  
!is.unsorted(before)
```

```
[1] TRUE
```

```
!is.unsorted(after)
```

```
[1] TRUE
```

Moreover, the values are sorted in increasing order within each group. Pairing them by row order would match the lowest “Before” value with the lowest “After” value, and so on, which would artificially create positive differences and bias a paired test.

For these reasons, the two sets of observations are treated as **independent groups**. This is a conservative choice: if the data were truly paired, a paired test would be more powerful, so we may miss a real effect.

2.2.3 Which test will you use? Parametric or non-parametric?

Hypotheses (two-sided test):

- H0: performance does not differ between before and after rehabilitation;
- H1: performance differs between before and after rehabilitation.

Although an increase is expected, a two-sided test is used, as it is more conservative and is the standard choice.

The choice of test depends on whether the observations are paired and on the distribution of the data:

Comparison	Parametric test	Non-parametric test
Paired data	Paired t-test	Wilcoxon signed-rank test
Independent data	Welch t-test	Mann–Whitney / Wilcoxon rank-sum test

Since the groups are treated as independent (question 2), the choice is between the Welch t-test and the Wilcoxon rank-sum test.

Normality is checked in each group with the Shapiro–Wilk test:

```
shapiro.test(before)
```

Shapiro-Wilk normality test

```
data:  before
W = 0.99683, p-value = 0.9999
```

```
shapiro.test(after)
```

Shapiro-Wilk normality test

```
data:  after
W = 0.98332, p-value = 0.9774
```

The Shapiro–Wilk test does not detect a significant deviation from normality in either group. However, with only 8 observations per group, the test has limited power.

Because of the small sample size, the data are mainly described using the median and interquartile range (IQR).

To remain consistent with this approach, a non-parametric Wilcoxon rank-sum test (Mann–Whitney test) will be used to compare performance before and after rehabilitation.

2.2.4 Which graph illustrates the effect of the treatment?

A boxplot compares the distribution of performance before and after rehabilitation. Because there are only 8 observations per group, the individual values are also displayed, and the mean of each group is shown for information (black diamond). The boxes show the median and interquartile range, which are the statistics compared by the main test (Wilcoxon rank-sum test). The groups are treated as independent, so the points are not connected between the two time points.

```
prepost %>%
  mutate(time = factor(time, levels = c("Before", "After"))) %>%
  ggplot(aes(x = time, y = perf, fill = time)) +
    geom_boxplot(alpha = 0.4, outlier.shape = NA) +
    geom_jitter(width = 0.08, size = 2) +
    stat_summary(fun = mean, geom = "point", shape = 18, size = 5) +
```

```
labs(x = "", y = "Performance (unit not specified)") +
theme_minimal() +
theme(legend.position = "none")
```

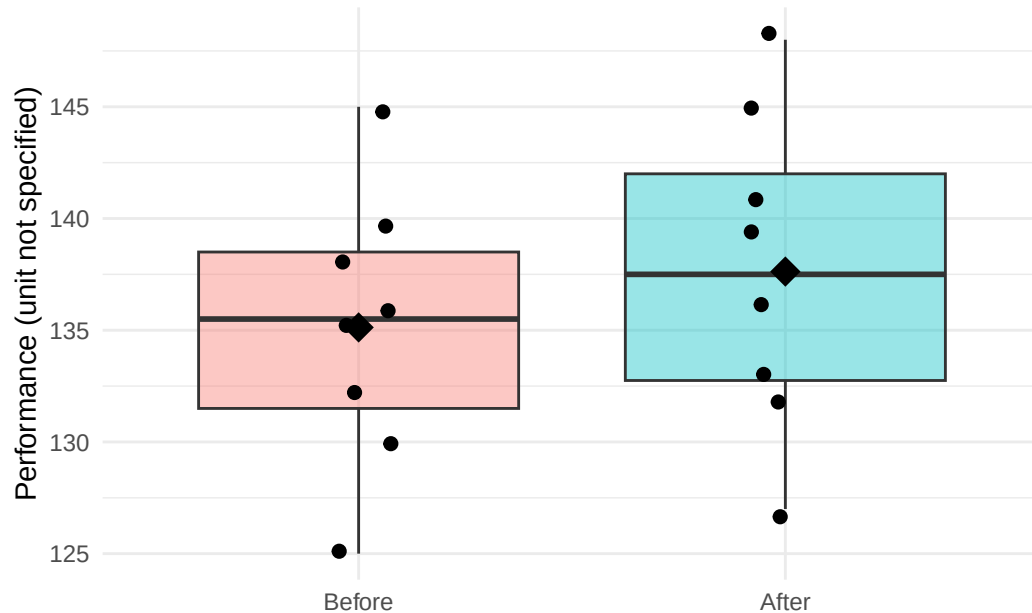


Figure 1: Physical performance before and after rehabilitation. Boxes show the median and interquartile range, dots the individual values, and black diamonds the group means.

The graph shows a slight upward shift of performance after rehabilitation (both the median and the mean are higher), but the two distributions largely overlap.

2.2.5 Statistical test and result

```
# main test: Wilcoxon rank-sum (Mann-Whitney) test, independent groups, two-sided
wilcox.test(after, before, paired = FALSE, exact = FALSE)
```

Wilcoxon rank sum test with continuity correction

data: after and before

```
W = 39.5, p-value = 0.4613
alternative hypothesis: true location shift is not equal to 0
```

```
# robustness check: Welch t-test (parametric equivalent)
t.test(after, before, paired = FALSE)
```

Welch Two Sample t-test

```
data: after and before
t = 0.75577, df = 13.793, p-value = 0.4625
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -4.604647  9.604647
sample estimates:
mean of x mean of y
 137.625   135.125
```

The Wilcoxon rank-sum test did not show a significant difference in performance between before and after rehabilitation ($W = 39.5$, $p = 0.46$). The Welch t-test, used as a robustness check, led to the same conclusion ($t = 0.76$, $df = 13.8$, $p = 0.46$), and the 95% confidence interval of the difference in means (-4.6 to $+9.6$) includes 0. Both tests therefore agree: the data do not show a significant effect of rehabilitation on performance.

Result sentence:

“Physical performance was not significantly different after rehabilitation compared with before (median: 137.5 [IQR = 9.25] vs 135.5 [IQR = 7.00]; Wilcoxon rank-sum test, $p = 0.46$).”

Limits. This non-significant result does not prove that the rehabilitation has no effect: it only means that the data do not provide sufficient evidence of an effect. Several factors limit the power of the analysis: (1) the very small sample size (8 observations per group); (2) the large variability between individuals compared with the observed difference; (3) the impossibility of using a paired design, because no participant identifier is available. If the same patients were measured before and after, a paired test, which removes the variability between individuals, would be more appropriate and more powerful. The conclusion could therefore change with properly identified paired data.

3 Testing some stereotypes

```
library(here)

snore <- read.table(
  here("data", "snore.txt"),
  header = TRUE
)
```

```
head(snore)
```

	age	weight	height	alcohol	sex	snore	tabac
1	47	71	158	0	H	N	0
2	56	58	164	7	H	O	N
3	46	116	208	3	H	N	0
4	70	96	186	3	H	N	0
5	51	91	195	2	H	O	0
6	46	98	188	0	F	N	N

3.1 Exploring the snore data

```
str(snore)
```

```
'data.frame': 100 obs. of 7 variables:
 $ age : int 47 56 46 70 51 46 40 46 49 39 ...
 $ weight: int 71 58 116 96 91 98 112 77 76 119 ...
 $ height: int 158 164 208 186 195 188 193 165 164 196 ...
 $ alcohol: int 0 7 3 3 2 0 5 0 0 3 ...
 $ sex : chr "H" "H" "H" "H" ...
 $ snore : chr "N" "O" "N" "N" ...
 $ tabac : chr "O" "N" "O" "O" ...
```

```
summary(snore)
```

age		weight		height		alcohol	
Min.	:23.00	Min.	: 42.00	Min.	:158.0	Min.	: 0.00
1st Qu.:	:43.00	1st Qu.:	: 77.00	1st Qu.:	:166.0	1st Qu.:	: 0.00
Median	:52.00	Median	: 95.00	Median	:186.0	Median	: 2.00
Mean	:52.27	Mean	: 90.41	Mean	:181.1	Mean	: 2.95
3rd Qu.:	:62.25	3rd Qu.:	:107.00	3rd Qu.:	:194.0	3rd Qu.:	: 4.25

```

Max.      :74.00   Max.      :120.00   Max.      :208.0   Max.      :15.00
      sex              snore              tabac
Length    :100   Length    :100   Length    :100
N.unique  : 2    N.unique  : 2    N.unique  : 2
N.blank   : 0    N.blank   : 0    N.blank   : 0
Min.nchar : 1    Min.nchar : 1    Min.nchar : 1
Max.nchar : 1    Max.nchar : 1    Max.nchar : 1

```

```
names(snore)
```

```
[1] "age"      "weight" "height" "alcohol" "sex"      "snore"  "tabac"
```

The dataset contains anthropometric measurements (age, weight, height) and categorical variables describing sex, smoking status, alcohol consumption and snoring status.

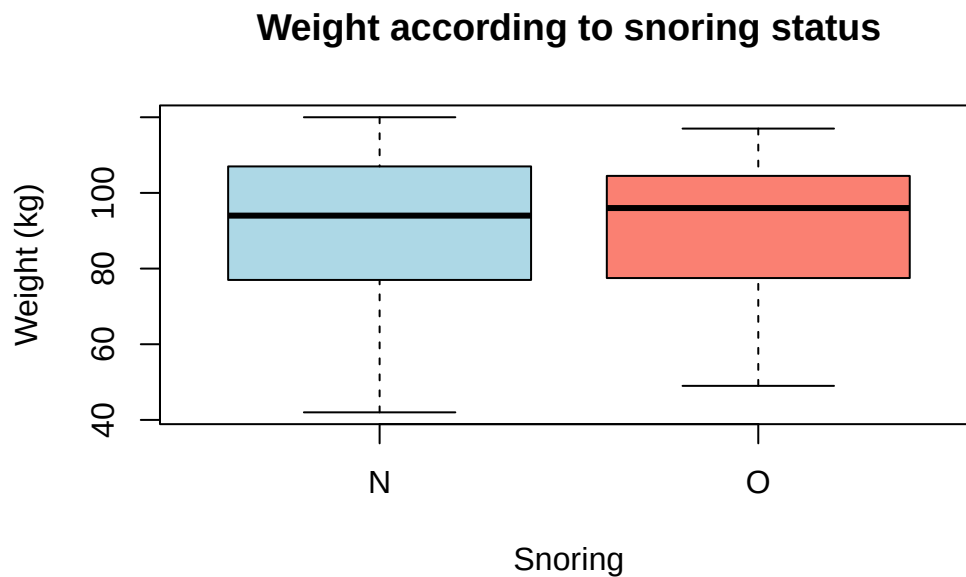
Statistical comparisons were performed to evaluate whether common stereotypes are supported by the data.

3.2 Are snorers fatter?

```

boxplot(
  weight ~ snore,
  data = snore,
  col = c("lightblue", "salmon"),
  xlab = "Snoring",
  ylab = "Weight (kg)",
  main = "Weight according to snoring status"
)

```



```
wilcox.test(weight ~ snore, data = snore)
```

Wilcoxon rank sum test with continuity correction

data: weight by snore

W = 1154, p-value = 0.9079

alternative hypothesis: true location shift is not equal to 0

3.2.1 conclusion

The data do not support the stereotype that snorers are heavier than non-snorers. A Wilcoxon rank-sum test showed no significant difference in weight between the two groups ($W = 1154$, $p = 0.908$).

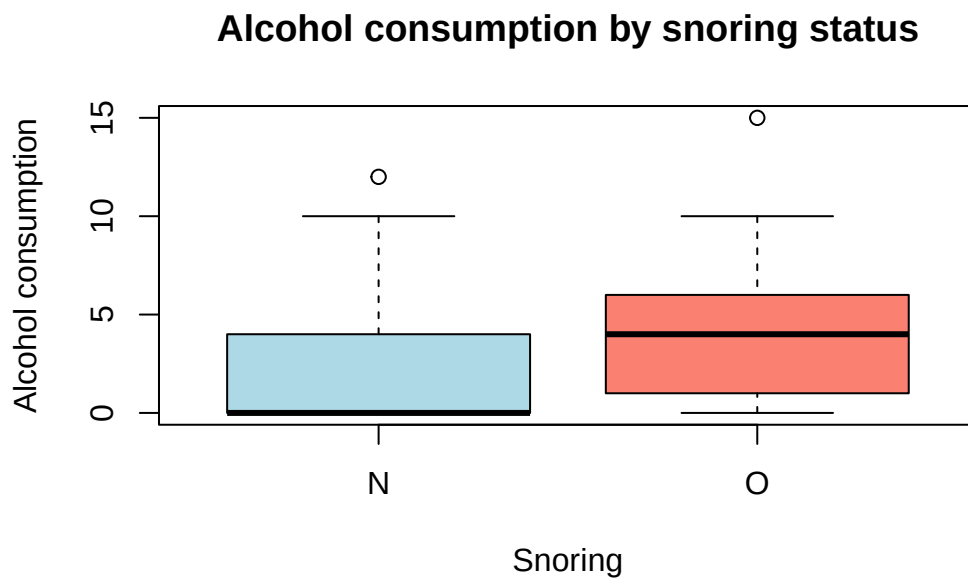
3.3 Do snorers drink more alcohol?

Let's compare alcohol consumption between snorers and non-snorers.

```

boxplot(
  alcool ~ snore,
  data = snore,
  col = c("lightblue", "salmon"),
  xlab = "Snoring",
  ylab = "Alcohol consumption",
  main = "Alcohol consumption by snoring status"
)

```



```

wilcox.test(alcool ~ snore, data = snore)

```

Wilcoxon rank sum test with continuity correction

data: alcool by snore

W = 788, p-value = 0.008594

alternative hypothesis: true location shift is not equal to 0

3.3.1 conclusion

Alcohol consumption was compared between snorers and non-snorers using a Wilcoxon rank-sum test. The test showed a statistically significant difference between the two groups ($W =$

788, $p = 0.009$). Therefore, the data support the stereotype that snorers consume more alcohol than non-snorers.

3.4 Do snorers smoke more?

```
table(snore$snore, snore$tabac)
```

```
      N  0  
N 21 44  
O 15 20
```

```
chisq.test(table(snore$snore, snore$tabac))
```

Pearson's Chi-squared test with Yates' continuity correction

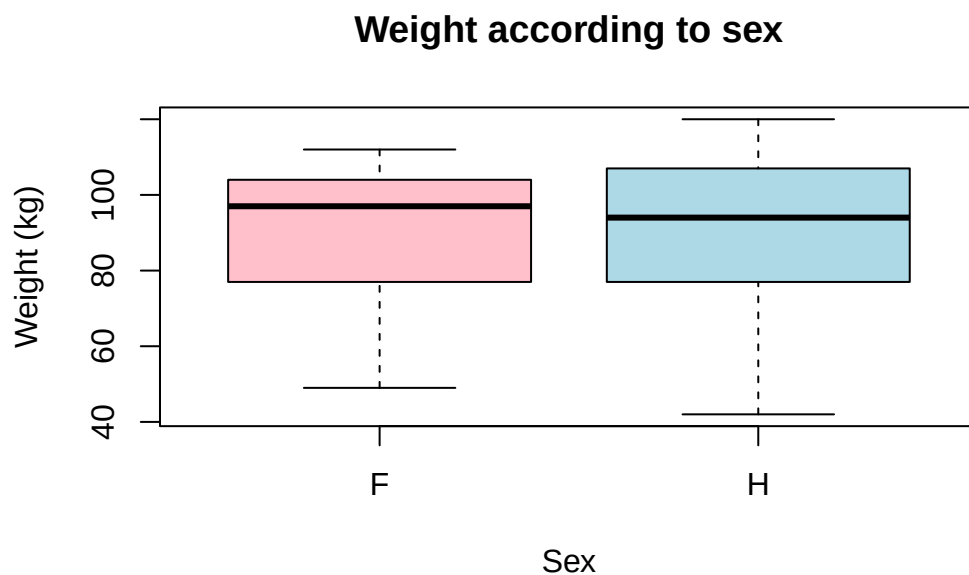
```
data:  table(snore$snore, snore$tabac)  
X-squared = 0.68872, df = 1, p-value = 0.4066
```

3.4.1 conclusion

The test showed no significant association between smoking and snoring status ($\chi^2 = 0.689$, $df = 1$, $p = 0.407$). Therefore, the data do not support the stereotype that snorers smoke more than non-snorers.

3.5 Are men fatter?

```
boxplot(  
weight ~ sex,  
data = snore,  
col = c("pink", "lightblue"),  
xlab = "Sex",  
ylab = "Weight (kg)",  
main = "Weight according to sex"  
)
```



```
wilcox.test(weight ~ sex, data = snore)
```

Wilcoxon rank sum test with continuity correction

data: weight by sex

W = 893.5, p-value = 0.729

alternative hypothesis: true location shift is not equal to 0

3.5.1 conclusion

The test showed no significant difference in weight between the two groups ($W = 893.5$, $p = 0.729$). Therefore, the data do not support the stereotype that men are fatter than women.

3.6 Do women smoke less?

```
table(snore$sex, snore$tabac)
```

```

      N    0
F 15 10
H 21 54

```

```
chisq.test(table(snore$sex, snore$tabac))
```

Pearson's Chi-squared test with Yates' continuity correction

```

data:  table(snore$sex, snore$tabac)
X-squared = 7.0023, df = 1, p-value = 0.00814

```

3.6.1 conclusion

The test revealed a significant association between sex and smoking status ($\chi^2 = 7.002$, $df = 1$, $p = 0.008$). Therefore, the data support the stereotype that smoking habits differ between men and women.

3.7 Are there any correlations between variables?

```

cor(
  snore[c("age", "weight", "height", "alcohol")]
)

```

	age	weight	height	alcohol
age	1.000000000	-0.003976806	0.01938144	-0.1587432
weight	-0.003976806	1.000000000	0.92697443	0.0915016
height	0.019381444	0.926974428	1.00000000	0.1063503
alcohol	-0.158743225	0.091501597	0.10635025	1.0000000

A strong positive correlation was observed between weight and height ($r = 0.927$), indicating that taller individuals generally weigh more. All other correlations were weak ($|r| < 0.20$), suggesting little or no linear relationship between the remaining variables.

3.8 Final Conclusion

This analysis examined several common stereotypes using statistical tests. The results showed that snorers did not weigh significantly more than non-snorers and were not more likely to smoke. However, snorers consumed significantly more alcohol than non-snorers. No significant difference in weight was found between men and women, whereas smoking status differed significantly between sexes, with women smoking less than men. Finally, correlation analysis revealed a strong positive relationship between height and weight, while the other quantitative variables showed only weak correlations.

Overall, the data support some stereotypes (higher alcohol consumption among snorers and lower smoking prevalence among women) but do not support others (snorers being fatter, snorers smoking more, or men being fatter).

4 Team workflow strategy with Git

4.1 Organisation of the repository

The repository `IEAP-Series02-RStudio` was created by Sarah Durand from RStudio (`usethis::use_git()` and `usethis::use_github()`), then Albatoul Ahmad and Damien Clanet were added as collaborators. The default branch was renamed from `master` to `main`, since `main` is the graded branch.

To allow three people to work on the same report without merge conflicts, we followed the strategy recommended in the assignment: a **master document** (`IEAP-Series02-Rstudio.qmd`) that only contains the YAML header and `{{< include >}}` statements, and **one sub-document per member** in the `Sections/` folder. Each member only edits their own file, so two people never modify the same file in parallel.

4.2 Branches and Pull Requests

- `main` always contains the validated version of the report.
- Each member works on a **personal branch** (e.g. `Sarah-2.2`, `albatoul-2.3`) and makes **at least one commit per question**, with a message describing the question answered (e.g. *“Answer 2.2.2 Q4: boxplot Before vs After”*).
- When a part is finished, it is merged into `main` through a **Pull Request** on GitHub (PR #1 for part 2.2, PR #2 for part 2.3).
- Before starting to work, each member pulls the latest version of `main`.

4.3 Issues

When a problem was detected, it was reported in a **GitHub Issue** and fixed in a dedicated branch. For example, issue #3 reported leftover YAML headers and dashes in the section files; it was fixed in the branch `fix-section-headers` and automatically closed by PR #4 thanks to the keyword `Closes #3`.

4.4 Final integration

The common parts of the report (this section, data sources, references and checklist) were added in the branch `final-report`, together with the final rendering of the PDF, and merged into `main` through a last Pull Request.

5 Challenges encountered and lessons learned

- **Setting up Git and GitHub.** Connecting RStudio to GitHub required creating a personal access token (`usethis::create_github_token()` and `gitcreds::gitcreds_set()`). *Lesson:* a token is a password and must never be shared; if it is exposed, it must be revoked immediately.
- **File paths in section files.** Running a chunk from a file in `Sections/` failed with “cannot open file ‘data/PrePost.csv’”, because the working directory is then the `Sections/` folder, whereas it is the project root when rendering the main document. *Lesson:* the `here` package builds paths from the project root and works in both cases.
- **YAML headers in included files.** RStudio adds a YAML header to every new Quarto document. Left in the section files, these headers (or their leftover `---` lines) hid titles or produced horizontal lines in the PDF. *Lesson:* included sub-documents must not have a header; the format is defined only in the master document.
- **Editors overwriting files.** After a `git pull`, a section file that was still open in an editor was saved again with its old (empty) content. *Lesson:* close or reload open files before pulling; Git made it easy to restore the correct version (`git checkout --<file>`).
- **Visual editor reformatting.** The RStudio visual editor reflowed some paragraphs, which produced large diffs without any content change. *Lesson:* use the Source mode to keep the Git history readable.
- **Respecting the workflow.** Part of the work on section 2.1 was pushed directly to `main` instead of going through a branch and a Pull Request. *Lesson:* agree on the workflow at the very beginning and protect the `main` branch if needed.

- **Interpreting imperfect data.** `PrePost.csv` contains no participant identifier and no unit, which prevented a paired analysis. *Lesson:* checking the structure of the data before choosing a statistical test is essential.

6 Data sources

- `PrePost.csv`: before/after performance of individuals with cardiac conditions following a rehabilitation training. Provided on the course Moodle: <https://moodle.umontpellier.fr/course/view.php?id=41022>
- `snore.txt`: anthropometric and lifestyle data (age, weight, height, alcohol, sex, snoring, smoking), one person per line. Provided on the course Moodle: <https://moodle.umontpellier.fr/course/view.php?id=41022>

7 References

7.1 Statistical methods

- Shapiro, S. S., & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika*, 52(3–4), 591–611. <https://doi.org/10.1093/biomet/52.3-4.591>
- Welch, B. L. (1947). The generalization of “Student’s” problem when several different population variances are involved. *Biometrika*, 34(1–2), 28–35. <https://doi.org/10.1093/biomet/34.1-2.28>
- Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6), 80–83. <https://doi.org/10.2307/3001968>
- Mann, H. B., & Whitney, D. R. (1947). On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics*, 18(1), 50–60. <https://doi.org/10.1214/aoms/1177730491>
- Pearson, K. (1900). On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling. *Philosophical Magazine*, 50(302), 157–175. <https://doi.org/10.1080/14786440009463897>

7.2 Software

- R Core Team (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>
- Wickham, H., et al. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43), 1686. <https://doi.org/10.21105/joss.01686>

- Müller, K. *here: A Simpler Way to Find Your Files*. R package. <https://CRAN.R-project.org/package=here>
- Allaire, J. J., et al. *Quarto*. <https://doi.org/10.5281/zenodo.5960048>

7.3 Part 2.1 – Statistical definitions

- Jamet, Mme. *Introduction aux statistiques* [lecture notes]. Licence STAPS, STAPS Marseille. Used for: variance and standard deviation, normal distribution, parametric and non-parametric tests, assumptions of parametric tests, p-value, risk of error, paired and unpaired tests.
- Google Gemini (including the AI-generated answers displayed by Google Search). Prompt (in French): “Explique-moi précisément ce que c’est que [name of the definition]” (e.g. “Explique-moi précisément ce que c’est que la médiane”). Accessed during the week of 28 September 2026. Used for: mean and median, assumptions of non-parametric tests. The answers were read and checked, then reformulated in English.

8 Checklist of the grading criteria

Repository

- ☒ Public GitHub repository named `IEAP-Series02-RStudio`
- ☒ `main` branch up to date (graded branch)
- ☒ Clear and complete `README.md`
- ☒ `LICENSE` file
- ☒ `IEAP-Series02-Rstudio.qmd` file
- ☒ Corresponding PDF file, rendered from the final version

Git history

- ☒ Clear history with meaningful commit messages
- [] At least one commit per question of the report (done for parts 2.2 and 2.3; part 2.1 was committed in a single commit, see *Challenges encountered and lessons learned*)
- ☐ Principled use of branches (one branch per member, Pull Requests into `main`) (done for parts 2.2, 2.3 and for the fixes and final report; part 2.1 was pushed directly to `main`, see *Challenges encountered and lessons learned*)

Content of the notebook

- ☒ Authors and date in the header
- ☒ Clickable link to the public GitHub repository
- ☒ Analysis of the team’s workflow strategy using Git
- ☒ Analysis of the challenges encountered and lessons learned

- ☒ Complete code and answers to all questions of the “Statistical tests” section (2.1, 2.2, 2.3)
- ☒ Data sources with their URLs
- ☒ Scientific references with their DOIs

Submission

- ☐ PDF uploaded on Moodle